

Total No. of Questions : 8]

SEAT No. :

[Total No. of Pages : 2

[6262]-35

T.E. (Computer Engineering)

DATABASE MANAGEMENT SYSTEMS

(2019 Pattern) (Semester - I) (310241)

Time :2½ Hours] | [Max. Marks : 70]

Instructions to the candidates:

- 1) Answer **Q.1** or **Q.2**, **Q.3** or **Q.4**, **Q.5** or **Q.6**, **Q.7** or **Q.8**.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Unit III --- Relational Database Design

Q1) a) What is functional dependency? Explain Armstrong's Axioms with suitable examples. [9]

b) Consider the relation R(A, B, C, D, E) with functional dependencies: FD1: A → BC

FD2: CD → E

FD3: B → D

FD4: E → A

i) Find all candidate keys for R. [9]

ii) Determine the highest normal form of R.

iii) Decompose R into 3NF if necessary.

OR

Q2) a) What is database normalization? Explain 1NF, 2NF, 3NF and BCNF with suitable examples for each. [8]

b) Consider the relation R(A, B, C, D, E, F) with functional dependencies: FD1: A → B

FD2: C → D

FD3: D → E

FD4: B → F

Check whether the decomposition of R into R1(A, B, F), R2(C, D, E) and R3(A, C) is lossless. Justify your answer. [9]

Unit IV --- Database Transaction Management

Q3) a) Explain ACID properties with suitable examples for each property. Draw the state diagram showing all possible states of a transaction. [9]

b) Consider the following schedule:

T1	T2	T3
R(A)		
	W(A)	

T1	T2	T3
R(B)		
		R(A)
	R(B)	
W(A)		
	W(B)	
		W(A)

Check whether the schedule is conflict serializable. If yes, give the equivalent serial schedule. [9]

OR

Q4) a) Explain timestamp-based concurrency control protocol. Discuss how read_TS and write_TS ensure serializability with an example. [9]

b) What is deadlock? Explain deadlock prevention, deadlock detection and deadlock avoidance techniques in detail. [9]

Unit V --- NoSQL Databases

Q5) a) Explain the CAP theorem. Discuss how different NoSQL databases make trade-offs between consistency, availability and partition tolerance. [9]

b) Compare RDBMS and NoSQL databases on at least 8 parameters. Explain the BASE properties of NoSQL databases. [9]

OR

Q6) a) Explain the different types of NoSQL data models: key-value store, document store, column-family store and graph database. Give one real-world example for each. [9]

b) Explain MongoDB CRUD operations with syntax and examples for: i) Insert operations (insertOne, insertMany) ii) Query operations (find with filters) iii) Update operations (updateOne, updateMany) iv) Delete operations (deleteOne, deleteMany) [8]

Unit VI --- Advances in Databases

Q7) a) What is semi-structured data? Explain the features of semi-structured data models. Compare JSON and XML as semi-structured data formats. [9]

b) Explain Object-Relational Database System. Discuss table inheritance and object-relational mapping (ORM) with examples. [9]

OR

Q8) a) Write short notes on any three: i) Active databases ii) Deductive databases iii) Spatial databases iv) Main memory databases [9]

b) Explain nested data types with respect to JSON and XML. Write a JSON document representing a university with departments, courses and students. [8]

This paper covers all four end-semester units with balanced weightage. Q1/Q2 test normalization depth (including candidate key computation), Q3/Q4 test transaction concepts with a real serializability check, Q5/Q6 cover NoSQL fundamentals with MongoDB syntax, and Q7/Q8 cover advanced database topics with a JSON writing task.